

Topology Ph.D. Exam
May 16, 2000

Work the following problems and show all work. Support all statements to the best of your ability. Work each problem on a separate sheet of paper.

1. Show that there is no retraction of the Möbius band onto its boundary.
2. State and prove the Contraction Mapping Theorem.
3. Give the Mayer-Vietoris sequence for homology. Use this to compute the homology groups for the n -sphere S^n .
4. Suppose that $f, g: X \rightarrow S^n$ are two maps such that $f(x)$ and $g(x)$ are not antipodal for all $x \in X$. Show that f and g are homotopic.
5. Show that no covering space of the 2-torus has the homotopy type of a figure 8.

Answer the following with complete definitions or statements or short proofs.

6. Give a classification of the compact surfaces.
7. State Urysohn's Lemma. State the Tietze Extension Theorem.
8. State the Five Lemma.
9. Define the mapping cylinder of $f: X \rightarrow Y$. Define the mapping torus of $f: X \rightarrow X$.
10. State the exact homology sequence for a pair of spaces.
11. State the Jordan Curve Theorem.
12. Define retraction and deformation retraction.
13. State the Ascoli (Arzela-Ascoli) Theorem.
14. Let V be a complete metric space such that no point is isolated. Can V be a countably infinite set?
15. Let A be a countable subset of $\mathbb{R}^2$. Show that $\mathbb{R}^2 \setminus A$ is arcwise connected.